# CS & DA




## Probability and Statistics

Statistics II

DPP- 01

Discussion Notes



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[NAT]

#Q. Consider two random variables X and Y with joint PMF given in Table.

| X \ Y | Y =2 | Y = 4 | Y = 5 | P(x) |
|---|---|---|---|---|
| X =1 | 1/12 | 1/24 | 1/24 | = 4/24 |
| X =2 | 1/6 | 1/12 | 1/8 | = 9/24 |
| X =3 | 1/4 | 1/8 | 1/12 | = 11/24 |
| P(y) | 6/12 | 6/24 | 6/24 | 1 |

| X | 1 | 2 | 3 | |
|---|---|---|---|---|
| P(x) | 1/6 | 3/8 | 11/24 | $\Sigma p_i = 1$ |

| Y | 2 | 4 | 5 | |
|---|---|---|---|---|
| P(y) | 1/2 | 1/4 | 1/4 | $\Sigma p_i = 1$ |

1) Find P ($X \le 2, Y \le 4$). $= P[X = 1 \text{ or } 2 \cap Y = 2 \text{ or } 4]$

$= P[(12) \text{ or } (14) \text{ or } (22) \text{ or } (24)]$

$= \frac{1}{12} + \frac{1}{24} + \frac{1}{6} + \frac{1}{12} = \frac{2+1+4+2}{24} = \frac{9}{24} = \frac{3}{8}$

**[NAT]**

2) Find the marginal PMFs of X and Y

$X =$ $Y:$

$P(x) =$ $P(y):$

**[NAT]**

3) Find P (Y=2| X =1) $= \frac{P(Y=2 \cap X=1)}{P(X=1)} = \frac{P(1,2)}{P(X=1)} = \frac{1/12}{1/6} = \frac{1}{2}$

[NAT]

4) Are X and Y independent?

$$P(X=x_i \cap Y=y_j) = P(X=x_i) \cdot P(Y=y_j) \quad \forall\, i \& j$$

eg $P(X=2, Y=2) = P(2,2) = \frac{1}{6}$

$P(X=2) \cdot P(Y=2) = \frac{3}{8} \cdot \frac{1}{2} = \frac{3}{16} \neq \frac{1}{6}$ So Not Ind

[NAT]

#Q. For the adjoining bivariate probability distribution of X and Y , find:

| X \ Y | Y =1 | Y = 2 | Y = 3 | Y =4 | Y =5 | Y =6 | P(X) |
|---|---|---|---|---|---|---|---|
| X =0 | 0 | 0 | 1/32 | 2/32 | 2/32 | 3/32 | 8/32 |
| X =1 | 1/16 | 1/16 | 1/8 | 1/8 | 1/8 | 1/8 | 10/16 |
| X =2 | 1/32 | 1/32 | 1/64 | 1/64 | 0 | 2/64 | 8/64 |
| P(Y) | 3/32 | 3/32 | 11/64 | 13/64 | 6/32 | 16/64 | |

1) $P(X \leq 1, Y=2)$ $= P(X=0 \text{ or } 1 \cap Y=2)$

$= P[(0,2) \text{ or } (1,2)]$

$= 0 + \frac{1}{16} = \frac{1}{16}$

2) $P(X \leq 1)$. $= P(x=0 \text{ or } 1) = \frac{8}{32} + \frac{10}{16} = \frac{8}{32} + \frac{20}{32} = \frac{28}{32}$

3) $P(Y \leq 3) = P(Y=1 \text{ or } 2 \text{ or } 3) = \frac{3}{32} + \frac{3}{32} + \frac{11}{64} = \frac{6+6+11}{64} = \frac{23}{64}$

| $Y$: | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $P(Y)$: | $\frac{3}{32}$ | $\frac{3}{32}$ | $\frac{11}{64}$ | $\frac{13}{64}$ | $\frac{6}{32}$ | $\frac{16}{64}$ |

4) $P(X<3, Y \leq 4)$. $= P(x=0 \text{ or } 1 \text{ or } 2 \cap y=1 \text{ or } 2 \text{ or } 3 \text{ or } 4)$

$= P(x=0, y \leq 4) + P(x=1, y \leq 4) + P(x=2, y \leq 4)$

$= \left(0+0+\frac{1}{32}+\frac{2}{32}\right) + \left(\frac{1}{16}+\frac{1}{16}+\frac{1}{8}+\frac{1}{8}\right) + \left(\frac{1}{32}+\frac{1}{32}+\frac{1}{64}+\frac{1}{64}\right) = \frac{9}{16}$

[NAT]

#Q. For the joint probability distribution of two random variables X and Y, find:

| X \ Y | Y =1 | Y = 2 | Y = 3 | Y =4 | Total |
|---|---|---|---|---|---|
| X =1 | 4/36 | 3/36 | 2/36 | 1/36 | 10/36 |
| X =2 | 1/36 | 3/36 | 3/36 | 2/36 | 9/36 |
| X =3 | 5/36 | 1/36 | 1/36 | 1/36 | 8/36 |
| X =4 | 1/36 | 2/36 | 1/36 | 5/36 | 9/36 |
| Total | 11/36 | 9/36 | 7/36 | 9/36 | 1 |

| X : | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| P(X): | 10/36 | 9/36 | 8/36 | 9/36 |

| Y : | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| P(Y): | 11/36 | 9/36 | 7/36 | 9/36 |

1)Find the marginal distributions of X amd Y

2) Conditional distribution of X given the value of Y =1 and that of Y given the value of X =2

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $P(X=x/Y=1)$ | $\frac{4}{11}$ | $\frac{1}{11}$ | $\frac{5}{11}$ | $\frac{1}{11}$ |

$$P(X=1/Y=1)=\frac{P(1,1)}{P(Y=1)}=\frac{4/36}{11/36}=\frac{4}{11}$$

$$P(X=2/Y=1)=\frac{P(2,1)}{11/36}=\frac{1/36}{11/36}=\frac{1}{11}$$

$$P(X=3/Y=1)=\frac{P(3,1)}{11/36}=\frac{5/36}{11/36}=\frac{5}{11}$$

$$P(X=4/Y=1)=\frac{4/36}{11/36}$$

| $y$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $P(Y=y/X=2)$ | $\frac{1}{9}$ | $\frac{3}{9}$ | $\frac{3}{9}$ | $\frac{2}{9}$ |

$$P(Y=1/X=2)=\frac{P(2,1)}{P(X=2)}=\frac{1/36}{9/36}=\frac{1}{9}$$

$$P(Y=2/X=2)=\frac{P(2,2)}{P(X=2)}=\frac{3/36}{9/36}=3/9$$

$$P(Y=3/X=2)=\frac{P(2,3)}{P(X=2)}=\frac{3/36}{9/36}=\frac{3}{9}$$

$$P(Y=4/X=2)=\frac{P(2,4)}{P(X=2)}=\frac{2/36}{9/36}=\frac{2}{9}$$

[NAT]

#Q. I have a bag containing 40 blue marbles and 60 red marbles. I choose 10 marbles (without replacement) at random. Let X be the number of blue marbles and y be the number of red marbles. Find the joint PMF of X and Y.

eg 3 Blue & 4 Red

& we want to take 3 Marbles w/o Replacement.

then Prob of drawing 1B & 2R are ?

Method I: (BRR) or (RBR) or (RRB)

$$\left(\frac{3}{7}\times\frac{4}{6}\times\frac{3}{5}\right)+\left(\frac{4}{7}\times\frac{3}{6}\times\frac{3}{5}\right)+\left(\frac{4}{7}\times\frac{3}{6}\times\frac{3}{5}\right)=\left(\frac{4\times3\times3}{7\times6\times5}\right)3$$

Method II:

$$\text{Req Prob}=\frac{{}^{3}C_{1}\times{}^{4}C_{2}}{{}^{7}C_{3}}=\frac{3\times\left(\frac{4\times3}{2}\right)}{\frac{7\times6\times5}{3\times2}}=\left(\frac{3\times4\times3}{7\times6\times5}\right)\times3$$

Fav Cases $=\{(0,10),(1,9),(2,8)\ldots(10,0)\}$

$x+y=10$.

$$P(x=x_i,y=y_j)=\begin{cases}\dfrac{{}^{40}C_{x_i}\ {}^{60}C_{y_j}}{{}^{100}C_{10}}, & x_i+y_j=10\\ 0, & \text{otherwise}\end{cases}$$

eg

$$P(x=1,y=2)=\frac{{}^{3}C_{1}\times{}^{4}C_{2}}{{}^{7}C_{3}}$$

[NAT]

#Q. The joint p.d.f of two random variables X and Y is given by:

$$f(x,y) = 9(1+x+y)/2((1+x)^4(1+y)^4) \quad 0 \le x < \infty, 0 < y < \infty$$

$$f(x) = \int_{-\infty}^{\infty} f(x,y)\,dy = \int_0^{\infty} \frac{9(1+x+y)}{2(1+x)^4(1+y)^4}\,dy = \frac{9}{2(1+x)^4}\int_0^{\infty} \frac{(1+y+x)}{(1+y)^4}\,dy = \frac{9}{2(1+x)^4}\int_0^{\infty}\left[(1+y)^{-3} + x(1+y)^{-4}\right]dy$$

Find the marginal distributions of X and Y , and the conditional distribution of Y for X=x.

$$= \frac{9}{2(1+x)^4}\left[\frac{(1+y)^{-2}}{-2} + x\cdot\frac{(1+y)^{-5}}{-5}\right]_0^{\infty} = \frac{9}{2(1+x)^4}\left[\frac{-1}{2(1+y)^2} - \frac{x}{5(1+y)^5}\right]_0^{\infty} = \frac{-9}{2(1+x)^4}\left[0 - \left\{\frac{1}{2} + \frac{x}{5}\right\}\right]$$

$$= \frac{3}{4}\cdot\frac{(3+2x)}{(1+x)^4}, \quad 0 \le x < \infty$$

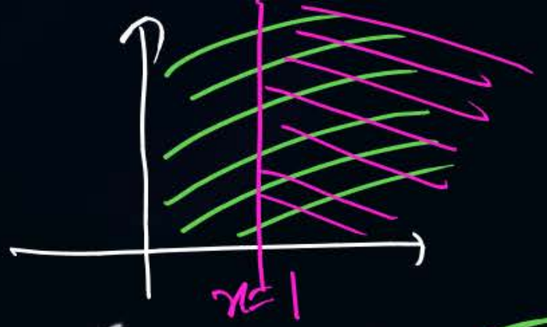
Similarly $f(y) = \int_0^{\infty} f(x,y)\,dx = \cdots = \frac{3}{4}\cdot\frac{(3+2y)}{(1+y)^4}, \quad 0 \le y < \infty$

$$P[Y=y/X=x] = \frac{f(x,y)}{f(x)} = \frac{\frac{9(1+x+y)}{2(1+x)^4(1+y)^4}}{\frac{3}{4}\frac{(3+2x)}{(1+x)^4}} = \frac{6(1+x+y)}{(1+y)^4(3+2x)}$$

$$0 \le y < \infty$$

**[NAT]**

#Q. Given-



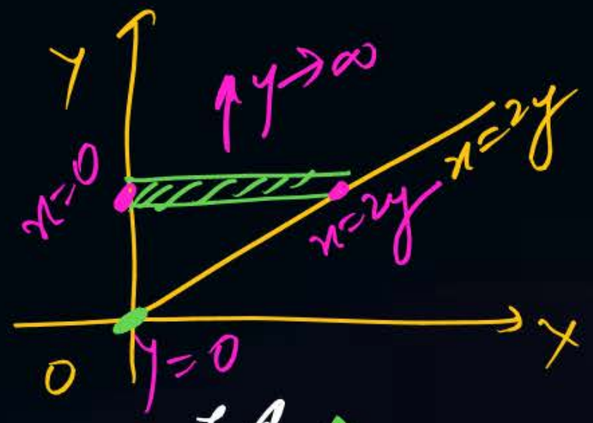

$$f(x,y) = \begin{cases} (e^{-(x+y)}), & 0 \le x < \infty, \; 0 \le y < \infty \end{cases}$$

Calculate the value of 1) $P(X>1) = \int_{y=0}^{\infty}\int_{1}^{\infty} f(x,y)\,dx\,dy = \int_{y=0}^{\infty}\int_{x=1}^{\infty} e^{-x}\cdot e^{-y}\,dx\,dy$

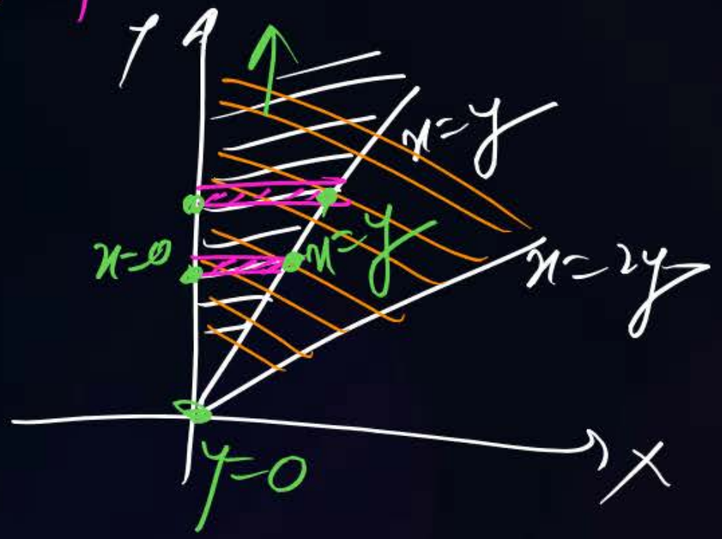
$$= \int_{x=1}^{\infty} e^{-x}dx \int_{y=0}^{\infty} e^{-y}dy = \frac{1}{e}\times 1 = e^{-1}$$

**[NAT]**

2) $P(X<Y|X<2Y) = \dfrac{P[x<y \cap x<2y]}{P(x<2y)} = \dfrac{P(x<y)}{\int_{y=0}^{\infty}\int_{x=0}^{2y} f(x,y)\,dx\,dy} = \dfrac{\int_{y=0}^{\infty}\int_{x=0}^{y} e^{-(x+y)}\,dx\,dy}{\int_{y=0}^{\infty}\int_{x=0}^{2y} e^{-(x+y)}\,dx\,dy}$



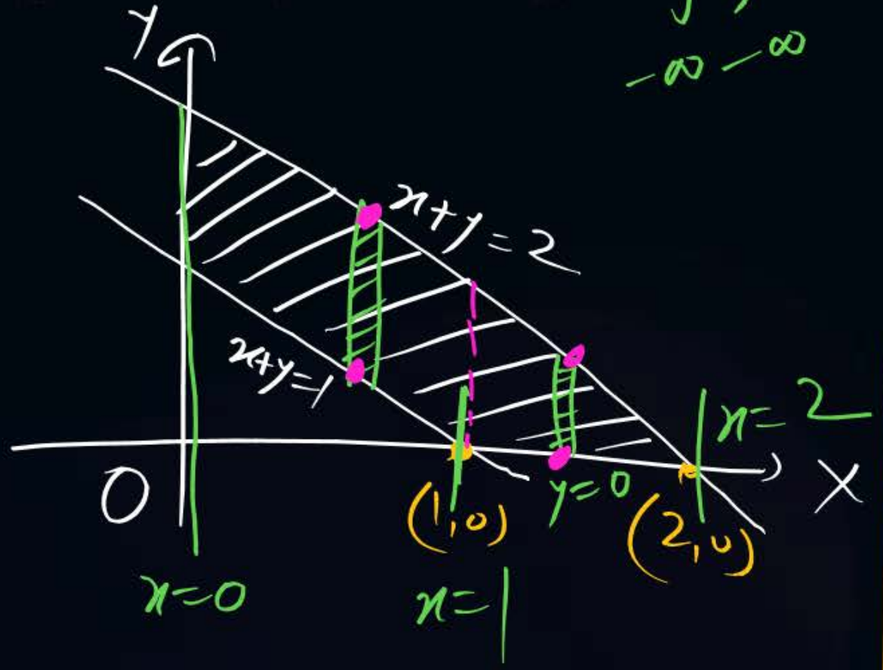

$= \dfrac{\int_{y=0}^{\infty}\left(\frac{e^{-x}}{-1}\right)_{x=0}^{y}(e^{-y}dy)}{\int_{y=0}^{\infty}\left(\frac{e^{-x}}{-1}\right)_{x=0}^{2y}(e^{-y}dy)} = \dfrac{\int_0^{\infty}(1-e^{-y})e^{-y}\,dy}{\int_0^{\infty}(1-e^{-2y})e^{-y}\,dy} = \dfrac{\int_0^{\infty}(e^{-y}-e^{-2y})\,dy}{\int_0^{\infty}(e^{-y}-e^{-3y})\,dy}$




$= \dfrac{1/2}{2/3} = \dfrac{3}{4}$ Ans

**[NAT]**

3) $P(1<X+Y<2) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = \int_{x=0}^{1}\int_{y=1-x}^{2-x} e^{-(x+y)}\,dy\,dx + \int_{x=1}^{2}\int_{y=0}^{2-x} e^{-(x+y)}\,dy\,dx$




$$= \int_{x=0}^{1}\left(\frac{e^{y}}{-1}\right)_{y=1-x}^{2-x}(e^{-x}dx) + \int_{x=1}^{2}\left(\frac{e^{y}}{-1}\right)_{0}^{2-x}(e^{-x}dx)$$

$$= \int_{0}^{1}\left(e^{x-1}-e^{x-2}\right)e^{-x}dx + \int_{x=1}^{2}\left(1-e^{x-2}\right)e^{-x}dx$$

$$= \int_{0}^{1}\left(e^{-1}-e^{-2}\right)dx + \int_{1}^{2}\left(e^{-x}-e^{-2}\right)dx$$

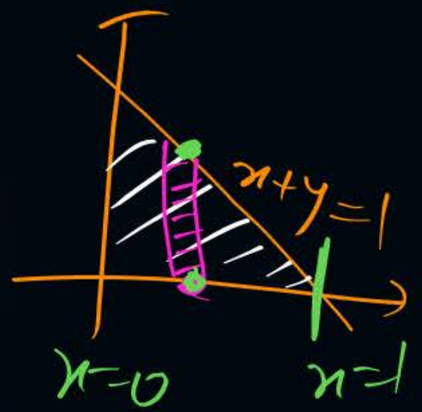
$$= \frac{2}{e} - \frac{3}{e^{2}}$$

[NAT]

#Q. Let X and Y be jointly continuous random variables with joint PDF

$$f_{X,Y}(x,y)=\begin{cases} cx+1 & x,y\geq 0, x+y<1 \\ 0 & \textit{otherwise} \end{cases}$$

1) Find the constant c.




w.k.t that $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$

$$\int_{x=0}^{1}\int_{y=0}^{y=1-x}(cx+1)\,dy\,dx = 1$$

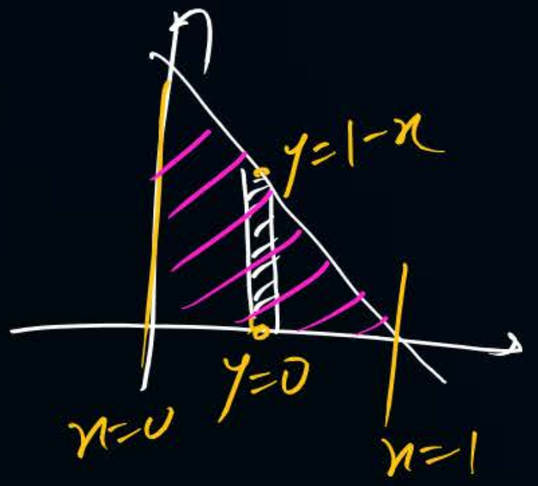
$$\int_{x=0}^{1}(1-x)(cx+1)\,dx = 1$$

$$c=3$$

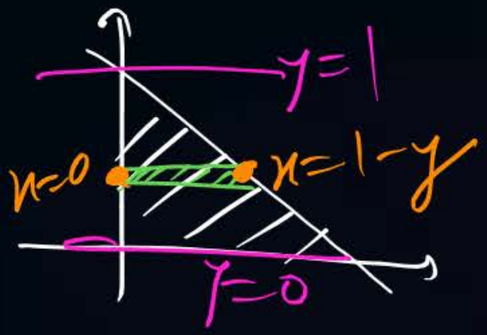
[NAT]

2) Find the marginal PDFs $f_X(x)$ and $f_Y(y)$.

$$f(x)=\int_{-\infty}^{\infty} f(x,y)\,dy=\int_{y=0}^{1-x}(3x+1)\,dy=(3x+1)(1-x),\quad 0\le x\le 1$$




(ii)



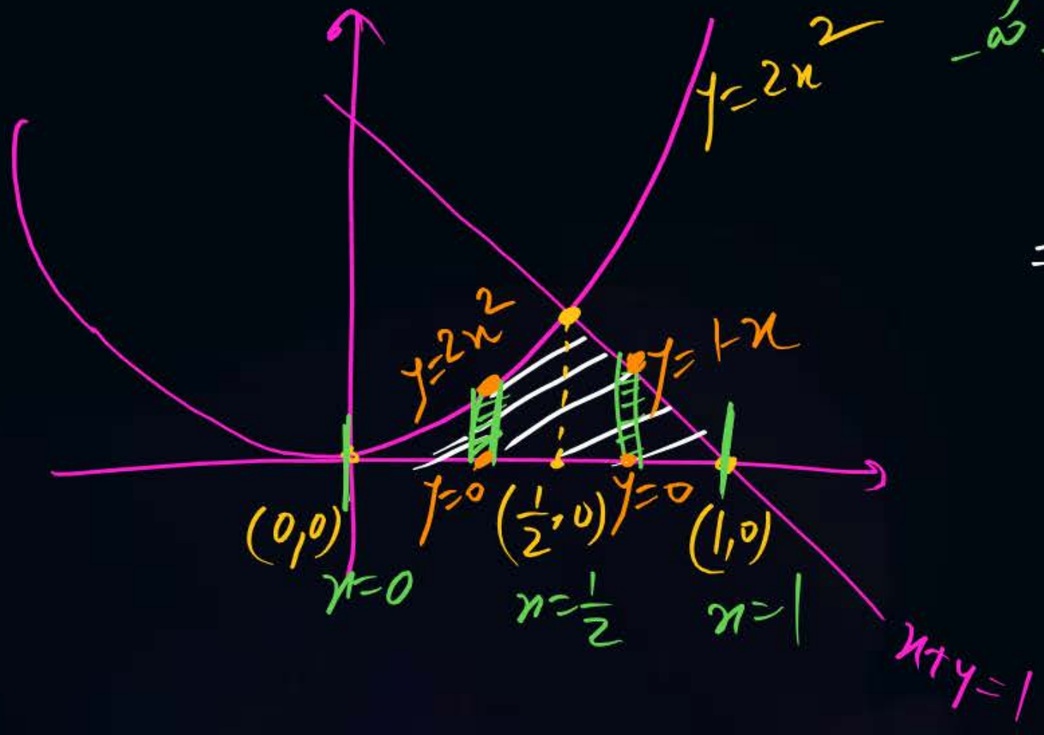

$$f(y)=\int_{-\infty}^{\infty} f(x,y)\,dx=\int_{x=0}^{1-y}(3x+1)\,dx=\left[3\left(\frac{x^2}{2}\right)+x\right]_0^{1-y}=\frac{3}{2}(1-y)^2+(1-y)$$

$$=\frac{1}{2}(1-y)(5-3y)$$

$$0\le y<\infty$$

**[NAT]**

3) Find $P(Y < 2X^2)$.




$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(3x+1)\,dx\,dy = \int_{x=0}^{\frac{1}{2}}\int_{y=0}^{2x}(3x+1)\,dy\,dx + \int_{x=\frac{1}{2}}^{1}\int_{y=0}^{1-x}(3x+1)\,dy\,dx$$

$$= \int_{x=0}^{\frac{1}{2}}(3x+1).(2x)\,dx + \int_{x=\frac{1}{2}}^{1}(3x+1).(1-x)\,dx$$

$$= 53/96$$

[NAT]

#Q. Let X and Y be jointly continuous random variables with joint PDF

$$f_{X,Y}(x,y)=\begin{cases}6e^{-(2x+3y)} & x,y\geq 0\\ 0 & \text{otherwise}\end{cases}$$

$f(x,y)=6e^{-2x-3y}$

$=(2e^{-2x}).(3e^{-3y})$

$=f(x)\,f(y)$

1) Are X and Y independent? YES

is these are Iid

E-Dist: $f(t)=\begin{cases}\mu e^{-\mu t}, & t\geq 0\\ 0, & \text{else}\end{cases}$

$\Rightarrow \mu=3$ & $E(t)=\frac{1}{\mu}=\frac{1}{3}$

Cross check: $f(x)=\int_{-\infty}^{\infty} f(x,y)\,dy=\int_{0}^{\infty} 6e^{-2x}e^{-3y}\,dy$

$=2e^{-2x}$ //

& $f(y)=3e^{-3y}, y\geq 0$

**[NAT]**

2) Find $E[Y|X>2] = E(Y) = \frac{1}{3}$

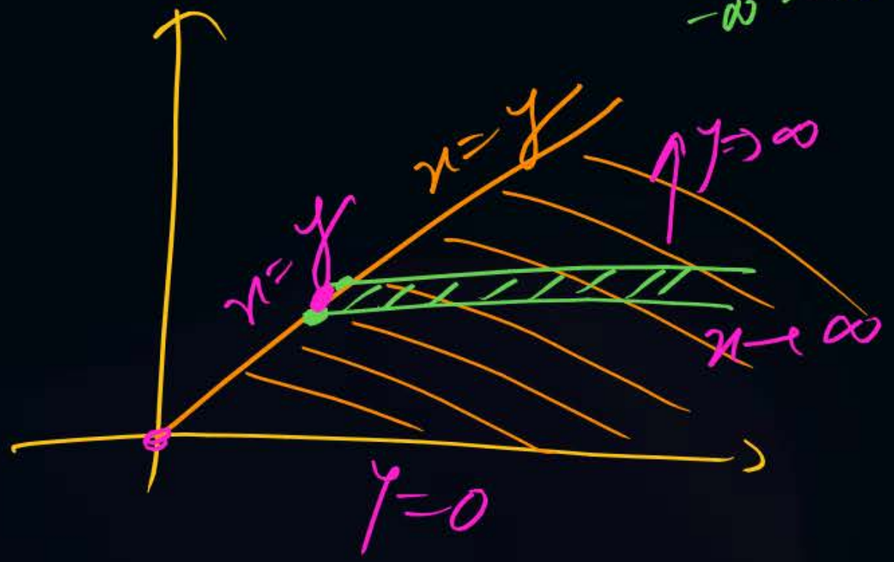
W.K that In Case of Independency, Condition has No significance.

Similarly $E(X) = \frac{1}{2}$

**[NAT]**

3) Find P (X >Y).

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = \int_{y=0}^{\infty}\int_{x=y}^{\infty} \left(6e^{-x}\cdot e^{-3y}\right)dx\,dy$$




$$= \int_{y=0}^{\infty} \left(\frac{e^{-2x}}{-2}\right)_{y}^{\infty} \left(6e^{-3y}\,dy\right)$$

$$= \int_{0}^{\infty} \frac{e^{-2y}}{2}\cdot 6e^{-3y}\,dy = 3\int_{0}^{\infty} e^{-5y}\,dy = \frac{3}{5}$$

[NAT]

#Q. Let X be a continuous random radiable with PDF

$$f_X(x) = \begin{cases} 2x & 0 \le x \le 1 \\ 0 & otherwise \end{cases}$$

$$f(y/x) = \begin{cases} \frac{1}{x-(-x)}, & -x \le y \le x \\ 0, & \text{otw} \end{cases}$$

Given X = x, the random variable Y is uniformly distributed on [−x, x].

1) Find the joint PDF $F_{XY}(x,y)$

i.e. $f(x,y) = \begin{cases} 1, & 0 \le x \le 1 \& -x \le y \le x \\ 0, & \text{otw} \end{cases}$

$$f(x,y) = \begin{cases} 1, & |y| \le x \le 1 \\ 0, & \text{otw} \end{cases}$$

w.k.t, $P(A \cap B) = P(A/B) \cdot P(B)$

$\Rightarrow f(x,y) = f(y/x) \cdot f(x)$

$= \frac{1}{2x} \cdot 2x = 1$

**[NAT]**

2) Find $f_Y(y)$. $= \int_{-\infty}^{\infty} f(x,y)\,dx = \int_{x=|y|}^{1} 9\,dx = 1-|y|$

$\therefore f(y) = \begin{cases} 1-|y|, & -1 \le y \le 1 \\ 0, & \text{otw} \end{cases}$

[NAT]

#Q. A bank operates a drive-up and a walk-up window. Here
X= the proportion of time the drive-up facility is in use
Y= the proportion of time the walk-up window in use
Say the manager has given the joint pdf based on his experience:

$$f(x,y)=\begin{cases}6/5(x+y^2) & 0\le x\le 1,\ 0\le y\le \frac{1}{4}\\ 0 & otherwise\end{cases}$$

The probability that neither facility is busy more than one-quarter of the time is?

$$P\left(0<x<\frac{1}{4},\ 0<y<\frac{1}{4}\right)=\int_0^{1/4}\int_0^{1/4} f(x,y)\,dx\,dy=\int_{y=0}^{1/4}\int_{x=0}^{1/4}\frac{6}{5}(x+y^2)\,dx\,dy$$

$$=\frac{6}{5}\int_{y=0}^{1/4}\left(\frac{x^2}{2}+xy^2\right)_{x=0}^{1/4}dy=\frac{6}{5}\int_{y=0}^{1/4}\left(\frac{1}{32}+\frac{y^2}{4}\right)dy=\frac{7}{640}$$

# THANK - YOU